Scanner Sequences

In this page you can find CBI's default sequences.

These are meant to be used as guidelines, but they can be modified to meet the needs of your study.

CBI Scout Sequences

Standard CBI fMRI Sequence

AP/PA Spin Echo Field Map

GRE Field Map

T1 MPRAGE Anatomical .9x.9x.9

Sagittal T2 .9x.9x.9

Standard DTI Sequence

DTI Spin Echo Field Map 2mm AP/PA

If you would like some advice on setting up your sequence, contact Chrysa Papadaniil.

If you want to double check what parameters were used for images already acquired, you can also look at the information from the header files or can ask us for a .pdf file of the sequence and we can send it to you.

Description: Automated alignment of slice positioning for head examinations. This option enables easy and accurate patient follow-up. AutoAlign references the 3D MR brain atlas and automatically aligns the slice positions in a standard reproducible way.	**Autoalign Scout:** Recommended.

Description: Standard Manual scout image. acquires 3 imaging planes and then manual placement of the following scan(s) are required.	**Standard Scout:**

The following sequence is used for both Resting state and for fMRI task runs. The Siemens system includes additional "dummy scans" at the beginning of the acquisition to allow the magnetization to stabilize to a steady state. The dummy scans are not stored, so they will make the banging sound like normal scans but there will be no data associated with them. There will be a trigger pulse that occurs with the first REAL SCAN (i.e. after the last dummy scan). This pulse will trigger your stim computer to start for timing purposes. The scanner will continue to deliver trigger pulses at the beginning of each subsequent volume.

Description	Standard EPI Sequence for fMRI
Acquisition Time	1300 ms per volume
Type	2D
Number of Slices	64
Slice Gap	0%
Bandwidth	2290 hz/px
Slice Thickness	2 mm
Multiband Accel. Factor	4
Geometry	Axial-autoalign set to head > brain
Slice order	Ascending inferior to superior / Mosaic
FOV	208 x 208
Resolution	2 mm x 2 mm
TR	1300 ms
TE	35 ms
Flip Angle	62 deg
Echo spacing	0.56 ms
Phase encoding	Anterior - Posterior

The below pair of Spin-Echo (SE) distortion map (a.k.a., field map) scans are used for both Resting state and fMRI task runs. Each one is acquired with opposite phase encoding (PE) directions.

Description	Spin Echo Field Map AP/PA. Required for the B0 correction to get preprocessed images.
Acquisition Time	27 seconds (3 volumes only) per PE direction
Type	2D
Number of Slices	64
Slice Gap	0%
Bandwidth	2004 hz/px
Slice Thickness	2 mm
Multiband Accel. Factor	1
Geometry	Make sure you "Copy Parameters" to match both the "Slices and Adjustment Volume" of the functional runs.
Slice order	Ascending inferior to superior
FOV	208 x 208
Resolution	2 mm x 2 mm
TR	6831 ms
TE	63.40 ms
Flip Angle	90 deg
Echo spacing	0.56 ms
Phase encoding	Anterior - Posterior (AP) or Posterior - Anterior (PA). The reversal of the phase encoding is achieved by enabling the "Reversed RO/PE" check box in the Sequence > Special tab.

If no CBI pre processing of images is wanted, users can acquire the standard GRE Fieldmap:

Description	Standard Siemens GRE Field map
Acquisition Time	1:42
Type	2D
Number of Slices	64
Slice Gap	0%
Slice Thickness	2 mm
FOV	208
Matrix	64 x 64
Resolution	2 mm x 2 mm
TR	635 ms
TEs	4.92 / 7.38 ms
Flip Angle	60 deg
Geometry	Make sure you "Copy Parameters" to match both the "Slices and Adjustment Volume" of the functional runs.

The T1 anatomical is collected on every new subject.

Description	T1 MPRAGE Hi res 3d structural
Acquisition Time	5 min 21 s
Type	3D
FOV	230 mm x 230 mm x 216 mm
Matrix	256 x 256 x 240
Resolution	0.9 mm isotropic
TR	2300 ms
TI	900 ms
TE	2.32 ms
Flip Angle	8 deg
Parallel imaging	GRAPPA Acceleration factor PE: 2 Reference lines PE: 24

The T2 scan is asked to be collected for our continuity for our BIDS format.

Description	T2 Mprage Hi res 3d structural
Acquisition Time	3 min 7 s
Type	3D
FOV	230 mm x 230 mm x 216 mm
Matrix	256 x 256 x 240
Resolution	0.9 mm isotropic
TR	3200 ms
TE	564 ms
Flip Angle Mode	T2 Variable
Parallel imaging	GRAPPA Acceleration factor PE: 2 Reference lines PE: 34
Turbo factor	314

Standard DTI Sequence

Description	B0 fieldmap correction scan for DTI
Acquisition Time	16 s
Type	2D
Slices	81
Slice gap	0
Slice thickness	2 mm
Slice order	Interleaved
FOV	220 mm x 220 mm
Matrix	110 x 110
Resolution	2 mm x 2 mm 6/8 fourier
TR	2720 ms
TE	54 ms
Diff directions	137
Diff weightings	1
b-value	0 s/mm^2
Multiband factor	3
Geometry	Make sure you "Copy Parameters" to match both the "Slices and Adjustment Volume" of the functional runs
Phase encoding	Anterior - Posterior (AP) or Posterior - Anterior (PA). The reversal of the phase encoding is achieved by enabling the "Reversed RO/PE" check box in the Sequence > Special tab.

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